

Does Cost-Efficiency Travel?

Comprehensive Research Report

All Findings, Analyses, Tests, and Artifacts

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Abstract

This document consolidates every analysis, finding, dataset, test, figure, and commit produced during the preparation of the AgenticOS @ NeurIPS 2026 submission. It is an internal reference, not a submission artifact.

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1 Executive Summary

Paper title: Does Cost-Efficiency Travel? What Agent Leaderboards Can Predict About Cost Across Workloads.

Venue: AgenticOS Workshop, NeurIPS 2026 (double-blind, 6 pages + appendix).

Core thesis: Every unconditioned aggregate we can compute from public agent data — raw cost ranks, success-adjusted efficiency, failure premiums — conflates which workloads were attempted with how agents behaved on them.

Three studies:

1. **Cost-rank portability:** Raw dollar-cost ranks transfer across HAL benchmarks (mean $\rho = 0.84$) but are absorbed by stable displayed-label cost propensity ($R^2 = 82.0\%$, residual ranks insignificant). CPS is less portable (mean $\rho = 0.59$).
2. **Scaffold burden:** At fixed task and model, OpenHands uses 15% fewer turns/tool calls than SWE-agent ($N = 977$ matched pairs, $p = 0.197$ for resolved status).
3. **Failure-burden:** The “expensive failure” premium is not detected under exact task matching: 107 discordant pairs, failed side heavier only 47–52/107 ($p \geq 0.25$), normalized ratio 0.98–1.06, MDE=15pp at 80% power.

Key commits:

- 884303e – telemetry analysis (TraceLab deep, MIMO deep, Exgentic)
- 96efb88 – complete telemetry appendix (MIMO replication, retry mechanism)
- 1a6af14 – failure-burden discordant-pair analysis
- c100d9a – output tables committed
- 1d33dd4 – integration into both papers with figure
- c0679bd – address five methodological weaknesses

2 Repository Structure

Path	Description
cost-efficiency-travel/	Main paper repo (this document)
efficient-benchmarking-ai-agents/	Ndzomga’s arXiv 2603.23749 (primary data source)
hal-paper-analysis/	Qualitative (Docent) + quantitative HAL analysis
hal-harness/	Official Princeton HAL harness (archived)
research_cache/	Reference papers: HAL, Efficient Benchmarking, SWE-Effi, Benchmark2
workspace/trace-audit/	Public trace dataset audit (cards + samples)
workspace/telemetry_audit/	Telemetry corpora (TraceLab DuckDB, MIMO 1,017 sessions, Exgentic)
hal_audit.py/.txt	First-pass HAL frontier/rank-correlation audit

3 Datasets

3.1 On hand

Dataset	What it is	Size	Status
HAL leaderboard CSV	242 evals, 9 benchmarks, 11 scaffolds, 29 labels	140 KB	Used in paper
Open-SWE-Traces	MiniMax-M2.5, OpenHands & SWE-agent, exact task/model match	851 MB	Used in paper
TraceLab DuckDB	8,058 real sessions, 665K rounds, 744K tool calls	160 MB	Used in paper
MIMO Claude Code	1,017 sessions, 4,690 rounds, one model	15 MB	Used in paper
Exgentic OTel	Multi-benchmark/harness, no rewards	50 MB	Audited only
Nebius SWE-rebench	Model trajectories + patches	5.6 GB	Audit card only
ITBench	105 SRE trajectories (IBM)	small	Audit card only
TRAIL	148 annotated error traces	gated	Not downloaded

3.2 Reference papers in `research_cache/`

- HAL: “Holistic Agent Leaderboard” (Kapoor et al., arXiv 2510.11977)
- Efficient Benchmarking: “Efficient Benchmarking of AI Agents” (Ndzomga, arXiv 2603.23749)
- SWE-Effi: “Re-Evaluating Software AI Agent System Effectiveness” (Fan et al., arXiv 2509.09853)
- Benchmark2: “Systematic Evaluation of LLM Benchmarks” (Qian et al., arXiv 2601.03986)

4 Primary HAL Study Results

4.1 Data and cohort

Frozen snapshot: 242 evaluations across 9 benchmarks, 11 scaffolds, 29 model labels. Fixed scaffold: HAL Generalist Agent. 6 benchmarks with ≥ 5 shared labels: CORE-Bench Hard, GAIA, SciCode, ScienceAgentBench, SWE-bench Verified Mini, TAU-bench Airline. Final cohort: 86 rows, 25 labels.

4.2 Cost-rank portability

- Raw dollar-cost Spearman ρ : 0.74–0.93 across high-overlap pairs (mean 0.84)
- All survive permutation + Holm correction
- Cost-propensity model: $R^2 = 82.0\%$ (adjusted 72.7%), 30 parameters, 56 residual df
- After removing label+benchmark fixed effects: residual cost ranks insignificant everywhere

4.3 Success-adjusted efficiency (CPS)

- Mean CPS $\rho = 0.59$ vs raw cost 0.84
- Survives permutation + Holm only for CORE–GAIA (0.87), CORE–TAU (0.83), GAIA–TAU (0.78)
- GAIA–SWE-mini: raw-vs-CPS gap 0.60 [0.14, 1.13]

4.4 LOBO held-out prediction

- Succeeds: CORE ($\rho = 0.78$), GAIA (0.87), SWE-mini (0.90), TAU (0.83)
- Fails: SciCode (-0.17), ScienceAgentBench (-0.64)

4.5 Frontier membership

No label appears on all 6 frontiers under any definition (discrete, hull, 5% tolerance). “No universal winner” is robust.

5 External Matched Trace Study

5.1 Scaffold burden ($N = 977$)

- OpenHands/SWE-agent median ratio: 0.85 for turns [0.82, 0.87], 0.85 for tool calls [0.82, 0.86]
- OpenHands lower on 69.7% of pairs
- Resolved status: no detected difference ($p = 0.197$)

5.2 Failure-burden ($N = 107$ discordant pairs)

Naive cross-task view (replicates folklore):

- OpenHands: failed/resolved turns ratio 1.21 [1.07, 1.36]
- SWE-agent: turns 1.19 [1.07, 1.35], characters 1.33 [1.11, 1.54]

Task-matched view (dissolves the premium):

- Sign test: failed side heavier 47–52/107 on every metric ($p \geq 0.25$)
- Scaffold-normalized ratio: 0.98–1.06, CIs span 1
- MDE at 80% power: $p1=0.65$ (15pp); observed 0.47–0.53 well below
- 95% CI [0.37, 0.57] raw, [0.43, 0.63] normalized

Power analysis (binomial test, $N = 107$, $\alpha = 0.05$):

- Critical values: reject if $k \leq 42$ or $k \geq 65$
- Power at $p1=0.55$: 0.14; $p1=0.60$: 0.48; $p1=0.65$: 0.85
- MDE at 80% power: $p1=0.65$

Scope: One model (MiniMax-M2.5), four-shard nonrandom sample, character/turn proxies not tokens/dollars. Rules out strong premiums but not modest ones.

6 Telemetry Appendix

6.1 TraceLab (8,058 sessions / 665,453 rounds)

Metric	Value
Median input / output tokens per round	132,092 / 249
Mean cache-read share	93.7%
Median context growth per session	47,267 tokens
Top 1% input-token share	48.3%
Next tool error after success / error	4.9% / 21.0% (4.29 $\times$)
Errored calls with same-tool retry / retry succeeds	88.3% / 77.9%
Consecutive-error run length, median / P90 / max	1 / 3 / 368

6.2 MIMO Claude Code (1,017 sessions / 4,690 rounds)

Metric	Value
Sessions with constant per-round usage	1,017 / 1,017 (100%)
Corpus totals (uncached / cache-read / output)	46.3M / 753.0M / 12.0M tokens
Cross-session concentration (top 10%)	37.9% of summed input
Error cascade: $P(\text{error} \mid \text{success}) / P(\text{error} \mid \text{error})$	1.0% / 4.3% (4.33 $\times$)

Cross-source replication: Error cascade ratios nearly identical: TraceLab 4.29 $\times$, MIMO 4.33 $\times$.

Data-quality audit: Per-round usage is constant within every MIMO session. Within-session dynamics cannot be validated. Source used only for session counts, corpus totals, cross-session concentration, and content-derived error sequences.

7 Figures and Tables

File	Content
figures/failure_burden_discordant.pdf	Two-panel: naive premium (left) vs discordant null (right)
figures/rank_transfer_compact.pdf	Pairwise rank transfer heatmap
figures/raw_cost_vs_cost_per_success.pdf	Raw-cost vs CPS rank scatter
figures/lobo_cost_portability.pdf	LOBO predicted vs observed ranks
figures/matched_trace_burden.pdf	Scaffold burden comparison (977 pairs)
figures/weak_pareto_membership.pdf	Frontier membership rates
figures/tracelab_tail_error_burden.pdf	Tail concentration + error burden
figures/coverage_heatmap.pdf	Label coverage across benchmarks
appendix/telemetry_anatomy.tex	Cross-source summary table
appendix/tracelab_key_metrics.tex + MIMO rows	TraceLab + MIMO key metrics

8 Test Suite

25 tests across 4 test files, all passing:

- `test_analysis.py` – rank correlation, bootstrap, Pareto frontier
- `test_portability.py` – CPS, LOBO, effect sizes
- `test_cps_inference.py` – permutation nulls, Holm correction
- `test_pipeline.py` – end-to-end pipeline hash check

Run command: `.venv/bin/python -m pytest -q`

9 Scripts

Script	Purpose
<code>run_analysis.py</code>	End-to-end HAL pipeline (rank correlations, Pareto, bootstrap)
<code>run_price_confound.py</code>	Log-cost two-way fixed-effects model
<code>run_aggregate_efficiency.py</code>	CPS and LOBO predictions
<code>run_external_trace_study.py</code>	Matched Open-SWE burden analysis
<code>run_reviewer_checks.py</code>	Permutation nulls, Holm, frontier sensitivity
<code>run_v2_robustness.py</code>	Leave-one-label-out, paired bootstrap
<code>run_tracelab_telemetry.py</code>	TraceLab DuckDB scale/growth/errors
<code>run_tracelab_deep.py</code>	TraceLab position/concentration/cascade
<code>run_tracelab_retry.py</code>	Tool-retry behavior after errors
<code>run_mimo_deep.py</code>	MIMO session-level summaries (audit-aware)
<code>run_failure_burden.py</code>	Failure-burden discordant-pair analysis
<code>build_paper_tables.py</code>	LaTeX table generation
<code>build_telemetry_figures.py</code>	Telemetry figures + tables
<code>build_failure_burden_figure.py</code>	Two-panel failure-burden figure
<code>build_compact_figure.py</code>	Compact rank-transfer figure

10 Commits (chronological)

Hash	Message
2dfa586	fix: package six-page AgenticOS submission
fd2f2ec	feat: add reviewer robustness analyses
b413072	feat: add matched external trace burden study
2dfa586	fix: package six-page AgenticOS submission
dfb9519	feat: test model cost-propensity confound
c3f2c16	docs: synchronize corrected paper artifacts
f9411bc	docs: add grounded practitioner efficiency vignette
f08fddb	feat: harden V2 robustness evidence
3f3229e	docs: restructure submission around portability decision flow
dacd787	docs: strengthen reviewer-facing positioning and safeguards
7018594	docs: rebuild reviewer-hardened paper PDFs
f3ad151	feat: add deep public telemetry audit
884303e	feat: add sequence and tail telemetry analysis
96efb88	feat: complete telemetry appendix with MIMO replication and retry mechanism
1a6af14	feat: failure-cost premium dissolves under exact task matching
c100d9a	chore: track failure-burden output tables
1d33dd4	feat: integrate failure-burden finding into both papers with figure
c0679bd	fix: address five methodological weaknesses

11 Remaining Pre-Upload Items

1. Re-check AgenticOS OpenReview portal fields/policy
2. Visual PDF review at 100% zoom
3. Supplement zip rebuild (predates telemetry/failure-burden additions)
4. Anonymization verification before upload
5. Deadline: August 29, 2026 AoE